INCH-POUND
MIL-V-24722/2(SH)
15 June 1990

MILITARY SPECIFICATION SHEET

VALVE, SERVO FLOW-CONTROL, ELECTROHYDRAULIC FOR 0 TO 10 GPM FLOW RANGE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index Specifications and Standards (DODISS) specified in the solicitation: MIL-V-24722.

1. GENERAL

1.1 Part or identifying number. Servovalves and subplates covered by this specification sheet shall be designated using Part or Identifying Numbers (PIN) as follows:

Example: M 24722 /2 −1

- M — Miltary specification with inch−pound units.
- 24722 — Specification number
- /2 — Specification sheet number
- −1 — Electrical receptacle style and location (see 1.3).

1.2 Mechanical configuration. The valve mounting pattern of this valve shall be as shown in figure 1. Envelope dimensions of the valve shall not exceed the envelope requirements of figure 2.

1.3 Electrical connector. Electrical receptacles for the −1 and −2 configuration and locations, as shown in figure 2, shall be in accordance with MS3102R14S-2P and MS3112H8-4P, respectively.

1.4 Torquemotor current. Rated differential current applied to the torquemotor for a maximum rated hydraulic flow shall be plus or minus 8 milliamperes (mA).

1.5 Torquemotor resistance. The measured pin-to-pin resistance shall be 1000 ± 15 ohms across each coil.

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2. TEST CONDITIONS

2.1 Test fluid. Hydraulic fluid for first article and quality conformance testing of valves conforming to this specification sheet shall be in accordance with MIL-L-17331, 2190 TEP.

2.2 Fluid contamination level. The valve shall perform in accordance with the first article and quality conformance test requirements specified herein when operated in a hydraulic system not cleaner than NAS 1638 class 8.

2.3 Set conditions. The valve shall be quality conformance tested within the table II specified limits for pressure, temperature, flow, and torquemotor signal command. Valve first article testing shall be performed in accordance with the limits and sequences of both tables I and II.

3. FIRST ARTICLE INSPECTION

The valve shall meet the first article inspection requirements specified in this section and in tables I and II.

4. QUALITY CONFORMANCE INSPECTION

The valve shall meet the quality conformance requirements specified in table II.

4.1 Null lap. No-flow cylinder port pressures shall be within 1500 $\pm$ 1000 pounds per square inch (lb/in^2) at valve null under the conditions specified in table II.

4.2 Hysteresis. The valve hysteresis shall not exceed plus or minus 3 percent of the maximum rated torquemotor current.

4.3 Threshold. Flow threshold shall not exceed plus or minus 0.25 percent of the maximum rated torquemotor current.

4.4 Pressure gain. The pressure gain shall meet or exceed 20 percent of maximum load pressure drop, as defined in SAE ARP 490, for each 1 percent change of signal input within the plus or minus 40 percent load pressure drop range.

4.5 Amplitude ratio. The valve amplitude ratio values shall be flat within plus or minus 3 decibels (dB) out to 60 hertz (Hz) when a sinusoidal torquemotor input command of not less than plus or minus 25 percent of rated current is applied.

4.6 Phase lag. The valve phase lag shall be less than 90 degrees at 60 Hz.

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TABLE I. *Required examination and test conditions – first article inspection.*

Test description	MIL-V-24722 requirements		M-24722/2 design and test require-ments	Test conditions[1]				Fluid flow[2] Set (S) Response (R)
				Fluid pressure (lb/in^2)		Fluid temp (°C)	Torque M current (percent)	
	Design	Test		Supply (P)	Return (R)			
GROUP 1								
Perform general examinations and tests of tables I and II of the general specification								
GROUP 2								
Perform examinations and tests per table II of this specification sheet								
GROUP 3								
Flow gain-low fluid temp	3.5.4.2	4.3.8	Fig 3	(P−R=1200±30)		32±1	Fig 3	(R)
Flow gain-hi fluid temp	3.5.4.2	4.3.8	Fig 3	(P−R=1200±30)		53 min	Fig 3	(R)
Magnetic field effects	3.5.2.9	4.3.7		(P−R=1200±30)		43±5	Fig 3	(R)
Endurance	3.5.6	4.3.20		3000±50	150±50	43±5	±100	10±0.25 gpm
Internal leakage[3]	3.5.3.6	4.3.11		3000±50	Open[4]	53 min	0	1180cc[5]
Flow gain[3]	3.5.4.2	4.3.8	Fig 3	(P−R=1200±30)		53 min	Fig 3	(R)
Proof pressure[3]	3.5.1.1.1	4.3.2.1		4500 min	Rsvr[6]	43±5	0	Zero
Shock	3.6.3	4.3.21						
Vibration	3.6.4	4.3.22						
Contam sensitivity	3.5.3.7	4.3.24	3.1	(P−R=1200±30)		43±5	±100	(S)

See footnotes on last page of table.

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TABLE I. *Required examination and test conditions – first article inspection – Continued.*

Test description	MIL-V-24722 requirements		M-24722/2 design and test require-ments	Test conditions[1]				Fluid flow[2] Set (S) Response (R)
				Fluid pressure (lb/in^2)		Fluid temp (°C)	Torque M current (percent)	
	Design	Test		Supply (P)	Return (R)			
Salt fog	3.6.5	4.3.25						
Maintenance	3.9	4.3.19						
GROUP 4								
Repeat examinations and tests per table II of this specification sheet								
GROUP 5								
Burst	3.5.1.1.3	4.3.28						
Test condition #1				7500 min	Rsvr	43±5	0	Zero
Test condition #2				4500 min	4500 min			

[1]Tests which have blank rows in the "Test conditions" columns are static, nonoperational tests which do not require a flow bench.
[2]Fluid flow is indicated to be either a set test condition (fixed or variable) or a dependent response with acceptance criteria specified in either the specification sheet (paragraph or figure) or the general specification.
[3]These tests shall be performed at 25, 50 and 75 percent intervals as specified in the endurance test requirements of the general specification.
[4]Open – Valve R port exhausts to atmosphere for fluid sample collection.
[5]Maximum allowable combined leakage from return port (R) after 2 10^6 cycles.
[6]Valve R port connected to test bench hydraulic reservoir.

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TABLE II. Required examination and test conditions – quality conformance (also first article test groups 2 and 4 as required).

Test description	MIL-V-24722 requirements		M-24722/2 design and test requirements	Test conditions[1]				Fluid flow[2] Set (S) Response (R)
				Fluid pressure (lb/in^2)		Fluid temp (°C)	Torque M current (percent)	
	Design	Test		Supply (P)	Return (R)			
Proof pressure	3.5.1.1.1	4.3.2.2	1.2					
Test condition #1				4500 min	Rsvr[3]	43±5	0	Zero
Test condition #2				3000 min	3000 min	43±5	0	Zero
External leakage	3.5.1.1.3	4.3.2.2		4500 min	4500 min	43±5	0	Zero
Polarity/run-in	3.5.4.1	4.3.3	1.3	3000±50	Rsvr	43±5	±100[2]	(S)
Internal leakage	3.5.3.6	4.3.11		3000±50	Open[4]	53 min	0	945cc[3]
Flow gain	3.5.4.2	4.3.8	Fig 3	(P–R=1200±30)		43±1	Fig 3	Fig 3
Rated flow	3.5.3.1	4.3.9		(P–R=1200±30)		43±2	±100	10±0.25 gpm
Null lap	3.5.4.4	4.3.12	4.1	3000±50	Rsvr	43±2	Pres null	C1 & C2 blocked
Hysteresis	3.5.4.5	4.3.13	4.2	3000±50	Rsvr	43±2	±100 var	Variable
Threshold	3.5.4.6	4.3.14	4.3	3000±50	Rsvr	43±2	Pres null ± 0.25	Zero

See footnotes on last page of table.

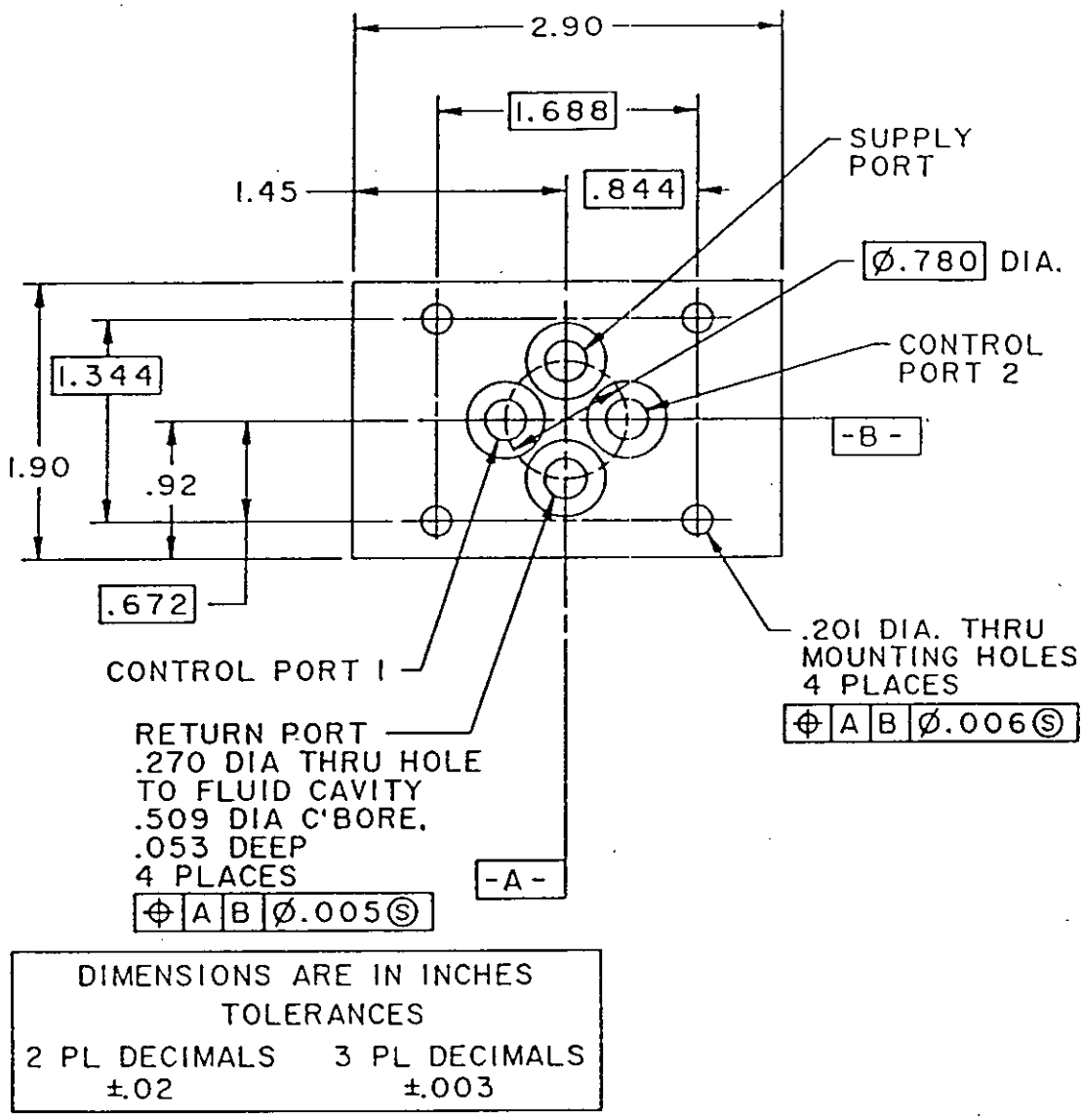
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TABLE II. Required examination and test conditions – quality conformance (also first article test groups 2 and 4 as required) – Continued.

Test description	MIL-V-24722 requirements		M-24722/2 design and test require-ments	Test conditions[1]				Fluid flow[2] Set (S) Response (R)
				Fluid pressure (lb/in^2)		Fluid temp (°C)	Torque M current (percent)	
	Design	Test		Supply (P)	Return (R)			
Pressure gain	3.5.4.7	4.3.15	4.4	3000±50	Rsvr	43±2	Pres null ±3.0	Zero
Amplitude ratio	3.5.5.1	4.3.16	4.5	3000±50	150±50	43±2	±25 min	±25%
Phase lag	3.5.5.2	4.3.17	4.6	3000±50	150±50	43±2	±25 min	±25%

[1]Tests which have blank rows in the "Test conditions" columns are static, nonoperational tests which do not require a flow bench.
[2]Fluid flow is indicated to be either a set test condition (fixed or variable) or a dependent response with acceptance criteria specified in either the specification sheet (paragraph or figure) or the general specification.
[3]Rsvr – Valve R port connected to test bench hydraulic reservoir.
[4]Open – Valve R port exhausts to atmosphere for fluid sample collection.

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NOTES:

1) VALVE MOUNTING SURFACE TO BE ⌓ .001
2) 32 SURFACE FINISH FOR C'BORE AND MOUNTING SURFACES
3) BREAK CORNERS APPROX .005 RAD

FIGURE 1. *Valve mouting surface configuration.*

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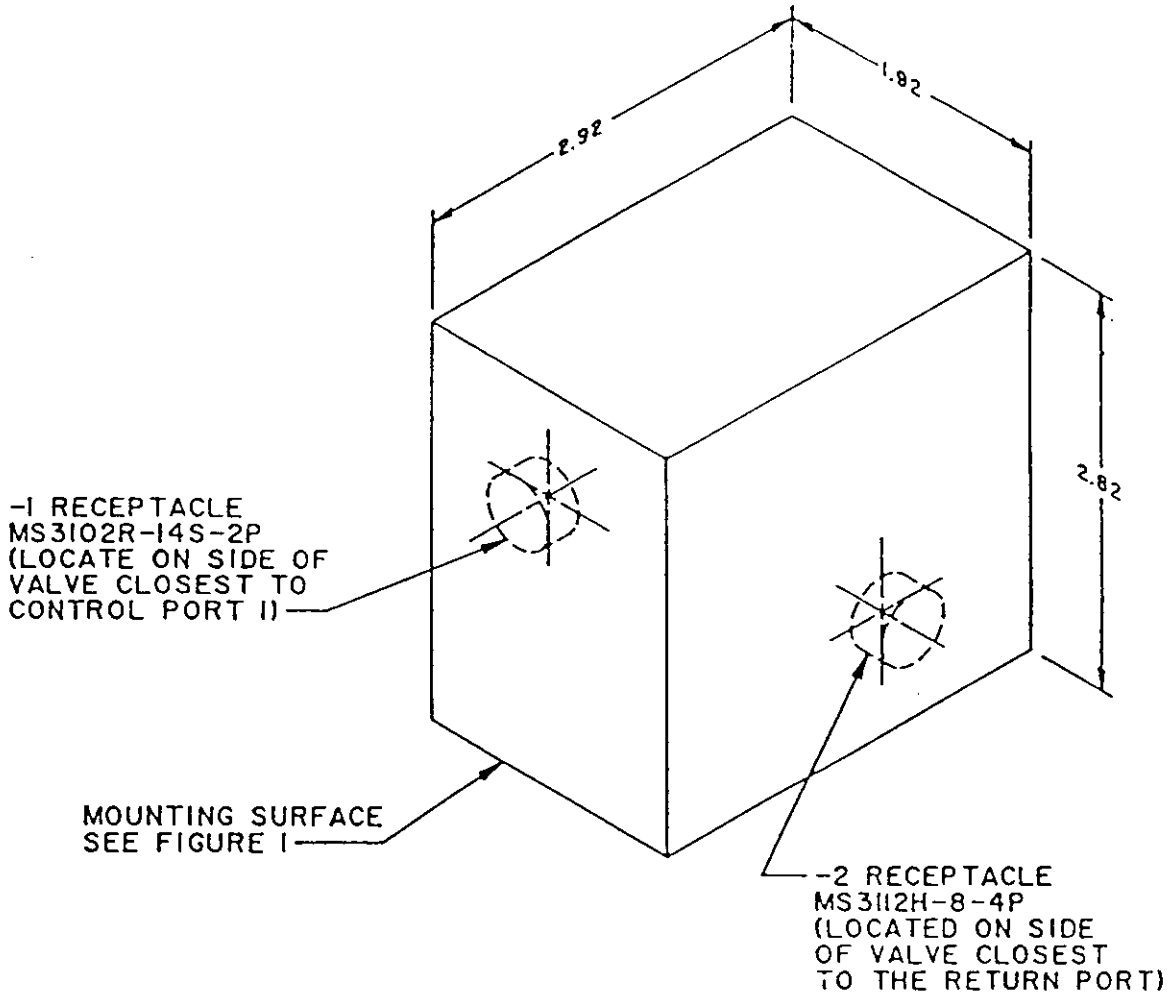


FIGURE 2. *Optional electrical receptacle locations and maximum valve envelope dimensions.*

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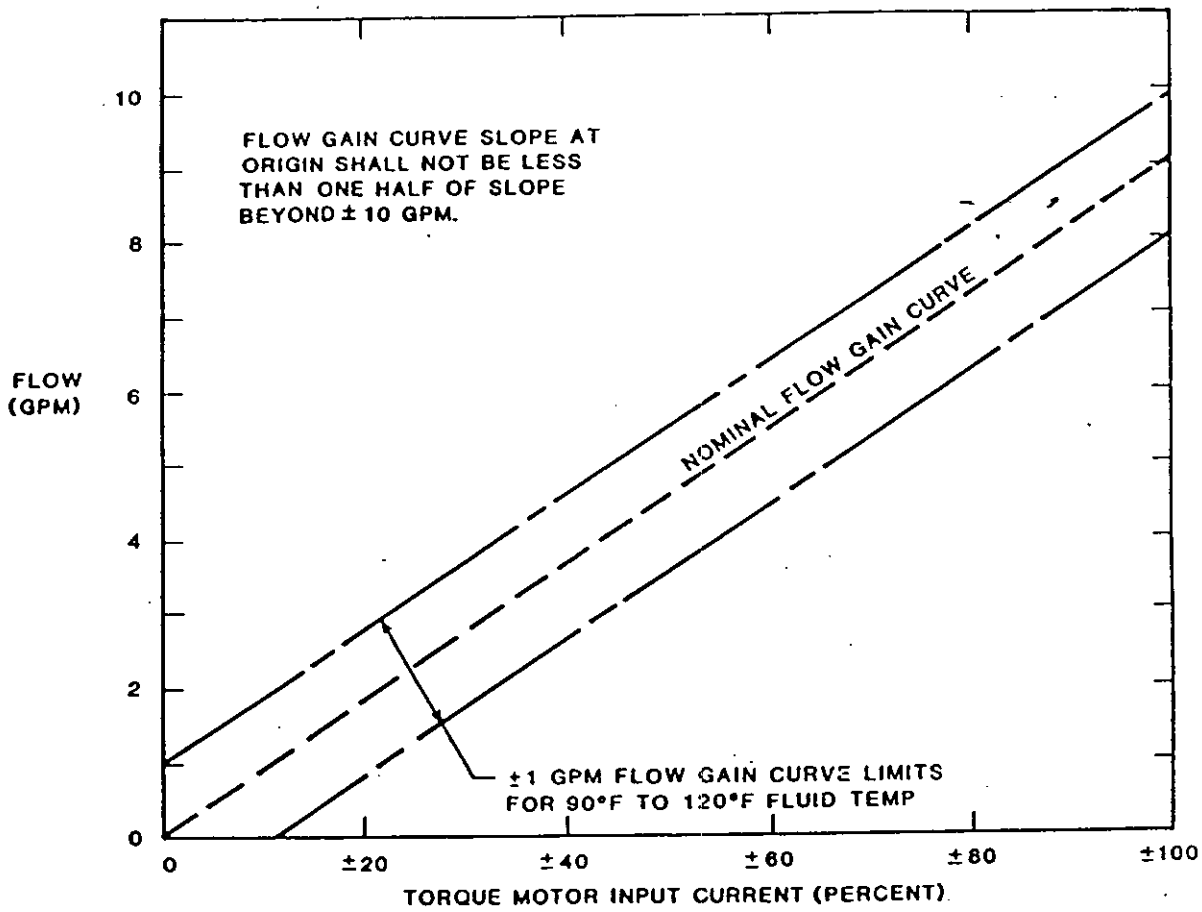


FIGURE 3. *Acceptable servovalve flow gain limits as a function of the fluid temperature.*

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